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Sex difference in ambulatory blood pressure control associates with risk of ESKD and death in CKD patients receiving stable nephrology care

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ABSTRACT

Background. It is unknown whether faster progression of chronic kidney disease (CKD) in men than in women relates to differences in ambulatory blood pressure (ABP) levels.

Methods. We prospectively evaluated 906 hypertensive CKD patients (553 men) regularly followed in renal clinics to compare men versus women in terms of ABP control [daytime <135/85 and nighttime blood pressure (BP) <120/70 mmHg] and risk of all-cause mortality and end-stage kidney disease (ESKD).

Results. Age, estimated glomerular filtration rate and use of renin–angiotensin system inhibitors were similar in men and women, while proteinuria was lower in women [0.30 g/24 h interquartile range (IQR) 0.10–1.00 versus 0.42 g/24 h, IQR 0.10–1.28, P = 0.025]. No sex-difference was detected in office BP levels; conversely, daytime and nighttime BP were higher in men (134 ± 17/78 ± 11 and 127 ± 19/70 ± 11 mmHg) than in women (131 ± 16/75 ± 11, P = 0.005/P < 0.001 and 123 ± 20/67 ± 12, P = 0.006/P < 0.001), with ABP goal achieved more frequently in women (39.1% versus 25.1%, P < 0.001). During a median follow-up of 10.7 years, 275 patients reached ESKD (60.7% men) and 245 died (62.4% men). Risks of ESKD and mortality (hazard ratio and 95% confidence interval), adjusted for demographic and clinical variables, were higher in men (1.34, 1.02–1.76 and 1.36, 1.02–1.83, respectively). Adjustment for office BP at goal did not modify this association. In contrast, adjustment for ABP at goal attenuated the increased risk in men for ESKD (1.29, 0.98–1.70) and death (1.31, 0.98–1.77). In the fully adjusted model, ABP at goal was associated with reduced risk of ESKD (0.49, 0.34–0.70) and death (0.59, 0.43–0.80). No interaction between sex and ABP at goal on the risk of ESKD and death was found, suggesting that ABP-driven risks are consistent in males and females.

Conclusions. Our study highlights that higher ABP significantly contributes to higher risks of ESKD and mortality in men.

Keywords: ambulatory blood pressure monitoring, chronic kidney disease, epidemiology and outcomes, sex

ADDITIONAL CONTENT

An author video to accompany this article is available at: https://academic.oup.com/ndt/pages/author_videos.

INTRODUCTION

Although women have a greater prevalence of chronic kidney disease (CKD), men are more often treated with dialysis [1]. The reasons for this discrepancy remain ill-defined and may potentially testify faster progression of renal disease in men, higher mortality rate in women or unequal access to renal replacement therapy [2, 3]. We recently demonstrated in CKD patients under nephrology care that men have higher risk of end-stage kidney disease (ESKD) and faster estimated glomerular filtration rate (eGFR) decline versus women, and that the higher renal risk in men was associated with higher proteinuria despite similar levels of blood pressure (BP) as detected by standard measurements in-office [4].

Previous studies analysing sex-related difference in CKD progression have shown conflicting results regarding hypertension

KEY LEARNING POINTS

What is already known about this subject?

- men with chronic kidney disease (CKD) have faster progression of CKD as compared with women;
- studies analysing sex-related difference in CKD progression have provided conflicting results on blood pressure (BP) control based on office measurement; and
- despite the fact that ambulatory BP (ABP) is superior to office BP in correctly diagnosing hypertension and predicting adverse outcome in CKD population, no study has addressed the impact of sex on the relationship between ABP and outcome in non-dialysis CKD population.

What this study adds?

- daytime and nighttime BP were significantly higher in men than in women, while office BP was similar. Consequently, men showed higher prevalence of masked hypertension while a white coat effect was prominent among women;
- ABP goal (daytime <135/85 and nighttime BP <120/70 mmHg) was achieved less frequently in men (25.1% versus 39.1%), while no sex-difference was detected in office BP at goal. Consequently, men showed higher prevalence of masked uncontrolled hypertension and sustained hypertension; and
- over 10.7 years of follow-up, men showed higher risk of end-stage kidney disease and all-cause mortality; this association was attenuated and become not significant after adjusting for the sex-difference in ABP control, while adjustment for office BP did not change results.

What impact this may have on practice or policy?

- this study highlights the importance of ABP monitoring as essential evaluation in men and women with CKD.

control [5–8]. Indeed, in the Chronic Renal Insufficiency Cohort (CRIC) Study, women had lower risk of CKD progression and death compared with men in the presence of similar office BP control rate [5]. Conversely, a patient-level meta-analysis reported a higher ESKD risk in women, whose systolic office BP was higher than in men (153 versus 147 mmHg) [6]. A secondary analysis of the Modification of Diet in Renal Disease (MDRD) study disclosed a slower eGFR decline in women versus men that was attenuated after accounting for higher mean BP and proteinuria at baseline [7]. The CKD Consortium reported a similar prevalence of hypertension in men and women without reporting BP values [8]. Finally, in the European QUALity Study on treatment in advanced CKD, office BP was similar in women and men with advanced CKD; in that study, men displayed a faster decline of renal function that was not affected by BP levels [9].

Of note, all the studies previously described defined hypertension control based on office BP measurements, while it is now well established that in CKD patients, hypertensive status is frequently misdiagnosed when based only on clinic BP [10–14]. Specifically, in patients under nephrology care, ambulatory BP (ABP) is more effective than office BP in predicting adverse outcomes (ESKD, mortality and cardiovascular events) [13–17]. Furthermore, ABP is useful for refining cardio-renal risk profile of CKD patients, being more strongly associated as compared with office BP with markers of cardiovascular disease and CKD progression, such as short-term BP variability and albuminuria [18–21].

We therefore performed a multicentre observational study to analyse the difference between office and ambulatory BP monitoring in women and men with non-dialysis CKD and the impact of in-office and out-of-office BP control on the sex-related difference in ESKD and mortality risk.

MATERIALS AND METHODS

We collected data on consecutive hypertensive CKD patients who underwent ABP monitoring during the period 2005–15 in three nephrology clinics in Italy. We enrolled patients with CKD Stages 2–5 (not on dialysis/transplant), hypertension (office BP ≥140/90 mmHg or antihypertensive treatment) and at least two visits to the nephrology clinic during a period ≥6 months. Exclusion criteria included acute changes in eGFR (either a decrease or an increase >30% in the previous 3 months), changes in antihypertensive therapy 2 weeks prior to ABP, inadequate ABP or atrial fibrillation. ABP monitoring was defined inadequate if reporting (i) <70% of expected measurements, (ii) <20 valid awake and <7 valid asleep measurements, and (iii) less than two valid daytime and one valid nighttime measurement per hour [22]. Institutional Review Boards of the participating centres approved the protocol and written informed consent was obtained from all patients.

Data on medical history, comorbidities, laboratory and current therapy were collected at baseline. In all participating centres, office BP was measured following standardized protocols as recommended by the European Society of Hypertension: the patient was in a quiet environment and BP was measured by trained staff in a sitting position three times at 5-min intervals using aneroid sphygmomanometers. Average of the six values of office BP recorded in the two consecutive days in which the ABP device was applied and removed was used for analysis.

All participating centres shared the same ABP protocols: Spacelabs 90207 monitors were used, with the cuff size (either 16×30, 16×36 or 16×42 cm) proportional to the patient's arm circumference and placed on the non-dominant arm; three BP readings were obtained concomitantly with clinic BP to ensure a difference <5 mmHg between the two sets of values. The monitor recorded BP every 15 min from 7:00 to 23:00 h and every 30 min from 23:00 to 7:00 h. Daytime and nighttime periods were derived from the patient's diary. Patients underwent ABP always on a workday, under regular antihypertensive treatment and without access to the ABP readings. At each centre, patient was always evaluated by the same nephrologist. Antihypertensive medications were prescribed to achieve office BP target and were administered from 8:00 to 22:00 h.

Patients were classified at goal for ABP when daytime and nighttime BP was <135/85 and <120/70 mmHg, respectively [22], and at goal for office BP if ≤140/90 mmHg if non-proteinuric and ≤130/80 mmHg if proteinuric [23]. Based on these thresholds and considering that 92% of patients were on antihypertensive treatment, we defined controlled BP if office and ambulatory BP were at goal, white coat effect (WCE) if only ABP was at goal, masked uncontrolled hypertension (MUCH) if only office BP was at goal and sustained hypertension (SH) if neither goal was met.

The aim of the study was to compare the BP profile (obtained by concomitant evaluation of ABP and office BP) between women and men and its impact on the risk of ESKD and all-cause mortality. ESKD was reached on the first day of chronic dialysis or pre-emptive kidney transplantation. Patients were followed up from the day of ABP (start date of follow-up) until 31 December 2018, ESKD or death before ESKD, and censored on the date they had the last clinic visit. If the patient missed a scheduled visit, he/she was called by the physician to set up a follow-up visit. If the cause of the missing visit was an event (ESKD or death), this information was verified and confirmed by means of either regional dialysis registry (date of ESKD) or death certificates provided by Mortality Registry Office.

Statistical analysis

Continuous variables are expressed as mean ± standard deviation (SD) or median [interquartile range (IQR)], and categorical variables as percentage. We used unpaired *t*-test, Mann–Whitney test or chi-square test when appropriate for comparisons between women and men. In the database, we had missing data for urinary sodium excretion ($n = 207$) and low-density lipoprotein cholesterol ($n = 344$). Since these variables were not included in survival models, we preferred not to use any imputation technique.

To estimate factors associated with ABP at goal in men and women, we used a multivariable logistic regression analysis adjusted for older age (>65 years), obesity [body mass index (BMI) >30 kg/m^2], current smoking, diabetes mellitus, history of cardiovascular disease, uncontrolled office BP, moderate to severe CKD (eGFR <45 mL/min/1.73 m^2), anaemia (haemoglobin <12 g/dL in women and <13 g/dL in men), severe proteinuria (24 h proteinuria >0.5 g/day) and use of more than two antihypertensive drugs (median value of the cohort). Sex difference in odds ratio was estimated for each variable by evaluating P-values obtained from the interaction terms between sex and each single variable included in the model. The same models were run separately in women and men to estimate odds ratio (OR) and 95% confidence interval (CI).

Because ESKD and death before ESKD are competing events—that is, occurrence of death prevents ESKD—we calculated the cumulative incidence of ESKD or death before ESKD by using the competing risk approach. Cumulative incidence curves were compared by using the Grey test [24]. The risks of the two endpoints were estimated by multivariable Cox proportional hazards models in which proportional hazards assumptions were evaluated using Schoenfeld residuals tests. We reported hazard ratio (HR) and 95% CI in four Cox models with progressive adjustment for sex (Model 1), sex and office BP at goal (Model 2), sex and ABP at goal (Model 3), and sex, office BP and ABP at goal (Model 4). All Cox models were stratified by Centre and CKD stage, and adjusted for potential confounders at baseline identified *a priori*: age, current smoking, BMI, diabetes, history of cardiovascular disease, left ventricular hypertrophy (LVH), haemoglobin, phosphate, 24 h proteinuria, eGFR and use of renin–angiotensin system (RAS) inhibitors. We used Cox models because the cause-specific relative hazards are more appropriate for studying the cause of disease in the case of a competing event [25].

Data were analysed using IBM SPSS version 26 (Armonk, NY, USA) and STATA, version 14 (Stata Corp.); two-tailed $P < 0.05$ was considered significant.

RESULTS

Out of 1116 eligible Caucasian patients, 906 subjects (353 women and 553 men) were included. Reasons for exclusion were inadequate ABP recordings ($n = 65$), change of antihypertensive therapy 2 weeks prior to the study ($n = 49$), atrial fibrillation ($n = 21$), eGFR change >30% ($n = 11$), lost after baseline visit ($n = 26$) and refusal to participate ($n = 38$).

Smoking habit and history of cardiovascular disease were more frequently recorded in men, while no difference was detected in age, BMI, diabetes and LVH (Table 1). Similarly, the cause of renal disease (hypertensive nephropathy 33%, diabetic kidney disease 22%, glomerulonephritis 18%, autosomal polycystic kidney disease 5% and other/unknown 22%) did not differ between the two groups ($P = 0.40$). eGFR was similar between women and men as well as distribution of CKD stages, while 24-h proteinuria was more severe in men in terms of absolute value and risk categories as well.

In the whole population, 31.3% of patients had office BP at goal and 30.6% had ABP at goal. As reported in Table 2, no sex difference was detected for systolic and diastolic office BP. Conversely, daytime and nighttime BP was higher in men. In particular, despite a similar number and type of antihypertensive drugs, daytime systolic and diastolic BP were on average 3.3 mmHg (95% CI 1.0–5.5) and 2.7 mmHg (95% CI 1.3–4.2) higher in men than in women; the same held true for nighttime systolic [3.7 mmHg (95% CI 1.1–6.3)] and diastolic BP [3.4 mmHg (95% CI 1.9–4.9)]. These differences translate into a

Table 1. Demographic, clinical and laboratory data of CKD patients stratified by gender

Parameters	Women ($N = 353$)	Men ($N = 553$)	P-value
Age (years)	63.3 ± 15.0	64.1 ± 13.4	0.41
BMI (kg/m^2)	28.2 ± 6.1	27.7 ± 4.2	0.11
Active smokers (%)	20.7	28.4	0.01
Diabetes (%)	24.6	27.5	0.34
History of cardiovascular disease (%)	21.8	33.5	<0.001
LVH (%)	65.4	69.4	0.21
CKD stages (%)			0.84
Stages 1–2	15.9	16.1	
Stage 3A	21.8	24.1	
Stage 3B	26.9	25.0	
Stage 4	24.1	25.1	
Stage 5	11.3	9.8	
eGFR (mL/min/1.73 m^2)	40.4 ± 22.4	41.4 ± 21.5	0.50
Haemoglobin (g/dL)	12.0 ± 1.5	13.0 ± 1.9	<0.001
Calcium (mg/dL)	9.39 ± 0.54	9.40 ± 0.58	0.97
Phosphorus (mg/dL)	4.11 ± 0.81	3.76 ± 0.70	<0.001
Cholesterol (mg/dL)	198 ± 45	187 ± 46	<0.001
Proteinuria (g/day)	0.3 (0.1–1.0)	0.4 (0.1–1.3)	0.02
Proteinuria categories			0.034
<0.15 g/day (%)	35.7	29.5	
0.15–0.50 g/day (%)	24.6	22.2	
>0.50 g/day (%)	39.7	48.3	

Data are mean ± SD, median (IQR) or percent. eGFR by CKD Epidemiology Collaboration equation.

Table 2. Office BP, ABP and treatment in CKD patients stratified by gender

Blood pressure	Women ($N = 353$)	Men ($N = 553$)	P-value
Office systolic BP (mmHg)	145 ± 21	144 ± 19	0.30
Office diastolic BP (mmHg)	81 ± 12	82 ± 11	0.31
24-h systolic BP (mmHg)	129 ± 17	132 ± 17	0.005
24-h diastolic BP (mmHg)	73 ± 11	75 ± 10	<0.001
Daytime systolic BP (mmHg)	131 ± 16	134 ± 17	0.005
Daytime diastolic BP (mmHg)	75 ± 11	78 ± 11	<0.001
Nighttime systolic BP (mmHg)	123 ± 20	127 ± 19	0.006
Nighttime diastolic BP (mmHg)	67 ± 12	70 ± 11	<0.001
Non-dipping (%)	66.3	72.7	0.04
Antihypertensive drugs (*n*)	2 (1–3)	2 (1–3)	0.33
RAS inhibitors (%)	72.8	70.5	0.46

Data are mean ± SD, median (IQR) or percent. Non-dipping defined by day/night ratio of systolic BP >0.9.

significantly lower control rate for daytime goal (46.3% versus 57.5%), nighttime goal (29.1% versus 43.3%) and both (25.1% versus 39.1%) in men versus women, while no sex difference was detected for office BP at goal (Figure 1A). Consequently, the distribution of pressor profiles obtained by combining office and ABP goals was different between sexes ($P < 0.001$), with

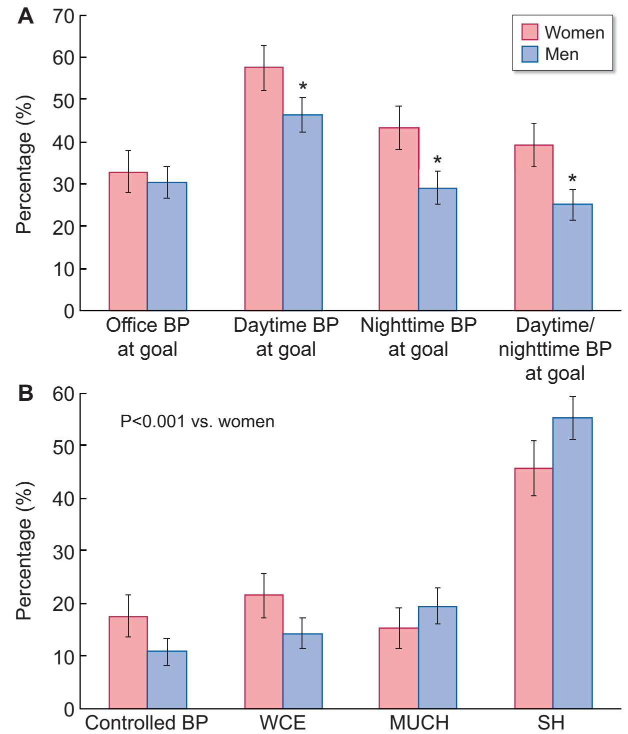

FIGURE 1: Achievement of BP goals with office and ambulatory monitoring (top) and distribution of altered pressor profiles (bottom) in CKD patients stratified by sex. *$P < 0.05$ versus women.

men showing higher prevalence of MUCH (19.5% versus 15.3%) and SH (55.3% versus 45.6%), and lower prevalence of WCE (14.3% versus 21.5%) (Figure 1B). Among women, the rate of ABP control did not differ between those aged <50 years (37.1%) and >50 years (39.6%).

Figure 2 depicts factors associated with ABP above goal, overall and by sex. As compared with women, adjusted odds of ABP not at goal were higher in men (OR 1.83, 95% CI 1.34–2.49). Furthermore, we found that smoking, office BP above goal, anaemia, moderate to severe CKD and proteinuria were associated with increased odds of having ABP above target in the whole population. However, only two factors increased the odds of ABP above target in women, moderate to severe CKD and anaemia (Figure 2).

Survival analysis

During a median follow-up of 10.7 years (IQR 5.0–13.5) after the initial visit, 275 patients progressed to ESKD (60.7% men) and 245 died (62.4% men). Cumulative incidence of ESKD was significantly lower in patients with ABP at goal, while incidence of death did not differ (Figure 3). No difference was found between men and women. However, after adjustment for demographic and clinical variables, we found that risk of ESKD was 34% higher in men and the risk of mortality was 36% higher in men (Model 1) (Table 3). These increased risks were unaffected by the adjustment for achievement of office BP at goal (Model 2). However, after adjusting the model for ABP at goal (Model 3), men showed an attenuated risk of adverse

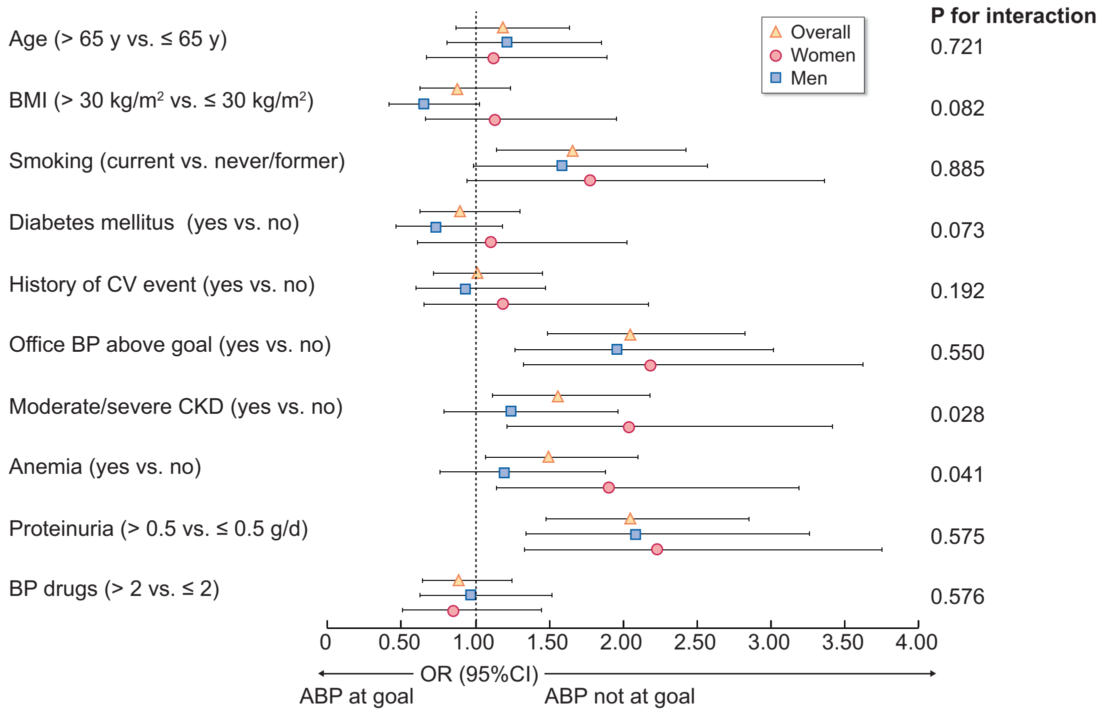

FIGURE 2: Multivariable logistic regression analysis evaluating demographic and clinical factors associated with achievement of ABP target overall and by sex. The analysis is adjusted by all the variables included. P-values indicate the significance of difference between women and men (interaction term).

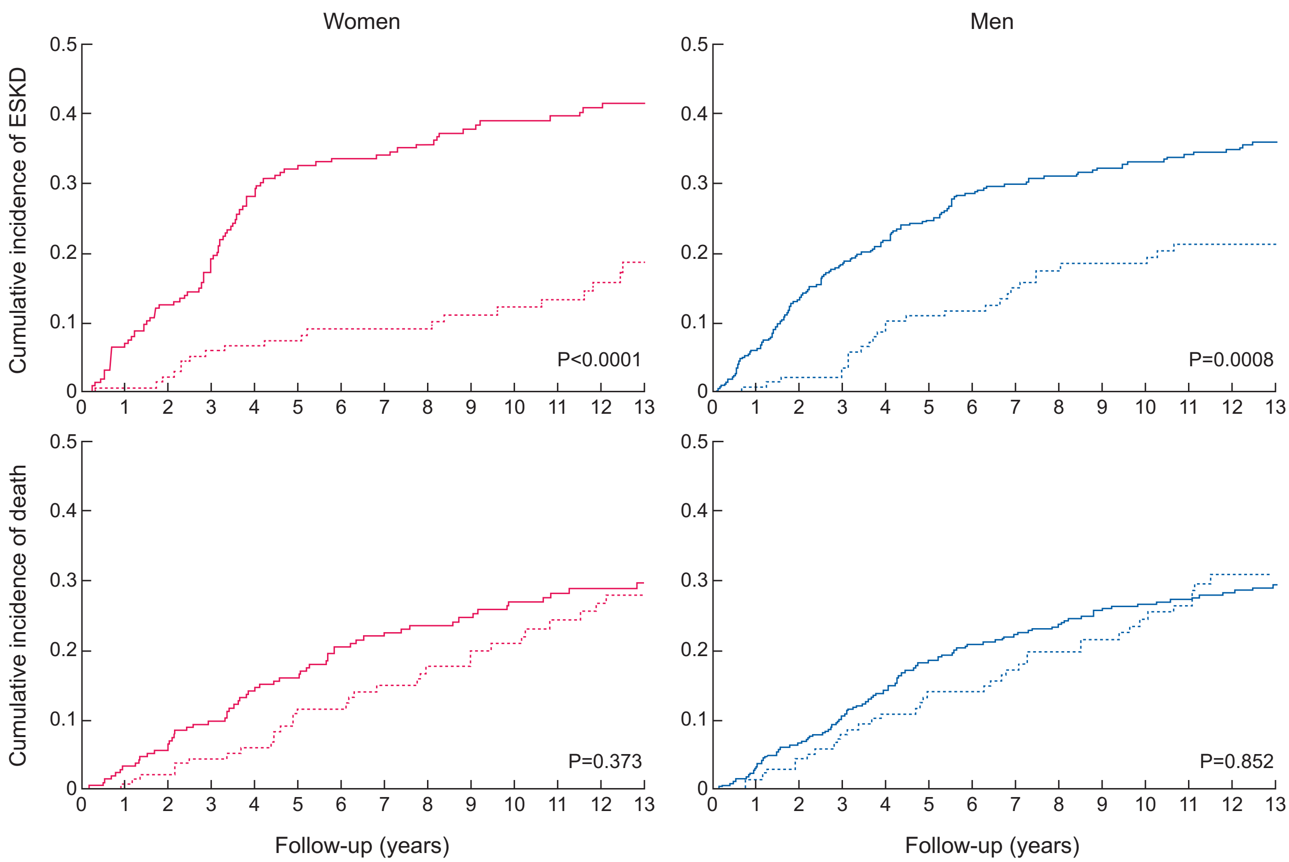

FIGURE 3: Cumulative incidence of ESKD and all-cause death by using competing-risk analysis in women and men stratified by ABP at goal (dotted line) and above goal (solid line).

Table 3. Adjusted cause-specific HR ratios for ESKD and all-cause death in women and men

	Risk of event [HR (95% CI)]			
	Model 1	Model 2	Model 3	Model 4
ESKD				
Men versus women	**1.34 (1.02–1.76)**	**1.35 (1.03–1.77)**	1.29 (0.98–1.70)	1.30 (0.98–1.71)
Office BP at goal	–	1.14 (0.86–1.51)	–	1.29 (0.96–1.73)
Ambulatory BP at goal	–	–	**0.52 (0.36–0.74)**	**0.49 (0.34–0.70)**
All-cause death				
Men versus women	**1.36 (1.02–1.83)**	**1.36 (1.02–1.83)**	1.31(0.98–1.77)	1.31 (0.97–1.77)
Office BP at goal	–	1.00 (0.73–1.35)	–	1.09 (0.80–1.49)
Ambulatory BP at goal	–	–	**0.60 (0.44–0.81)**	**0.59 (0.43–0.80)**

Model 1 is stratified by CKD stage and center and adjusted for age, smoking, BMI, diabetes mellitus, history of cardiovascular disease, LVH, phosphate, haemoglobin, eGFR, proteinuria, number of antihypertensive drugs and use of inhibitors of RAS.
Model 2 adjusted as Model 1 + office BP at goal ($\leq$140/90 mmHg if non-proteinuric and $\leq$130/80 mmHg if proteinuric).
Model 3 adjusted as Model 1 + ABP at goal (daytime $<$135/85 mmHg and nighttime $<$120/70 mmHg).
Model 4 adjusted as Model 1 + office and ABP at goal.
Significant hazard ratios are indicated in bold.

outcomes. This was emphasized in the fully adjusted model (Model 4); indeed, accounting for both office BP and ABP at goal, men were not exposed at higher risk for ESKD (HR 1.30, 95% CI 0.98–1.71) and all-cause mortality (HR 1.31, 95% CI 0.97–1.77). The absence of interaction between sex and ABP at goal on either ESKD (P = 0.30) or death risk (P = 0.75) indicated that the association between ABP and outcome is consistent in men and women. Interestingly, office BP at goal did not associate with reduced risk of ESKD or death whereas ABP at goal was associated with a risk of ESKD and mortality that was 51% and 41% lower, respectively (Table 3).

DISCUSSION

Our study provides first-time evidence in non-dialysis CKD patients under stable nephrology care that ABP profile differs between men and women, with men showing higher BP during the day and night and, more important, that sex difference in ABP control significantly contributes to the poor prognosis of men compared with women. Notably, ABP and survival differences were disclosed in the absence of any difference in office BP analysed as either continuous or categorical (BP at goal) variable.

An original finding of this study is the evaluation of factors associated with uncontrolled ABP in men and women. We found that in our population when considered as a whole, current smoking, uncontrolled office BP, more advanced renal damage (eGFR $<$45 mL/min/1.73 m^2 and severe proteinuria) and anaemia were all associated with ABP above target (Figure 2). Therefore, in the presence of one or more of these conditions, it is likely that patients may have ABP above goal and, therefore, either MUCH or SH. Interestingly, this association with either lower eGFR or anaemia differed by sex, being significantly associated with uncontrolled ABP in women but not in men. Agarwal and Andersen previously showed in a cohort of veterans with CKD that albuminuria, race and number of BP medications were predictors of higher 24 h systolic BP [26]. In that study, however, the role of sex was not evaluated since the cohort included almost exclusively men (96%). The International Database of Ambulatory BP in Renal Patients (I-DARE) study, including 7518 patients with non-dialysis CKD from the USA, Italy, Spain and Japan, reported data consistent with our findings [11]. Indeed, this collaborative study found that male sex, eGFR $<$30 mL/min/1.73 m^2 and treatment with a higher number of BP drugs were associated with uncontrolled ABP independent of office BP control [11].

Previous reports have shown that men have a faster progression of renal disease and higher risk of both ESKD and mortality [3–5, 9, 27]. In some but not all of these studies, men had similar or even lower office BP levels in comparison with women, a finding that may apparently exclude hypertension as critical mechanism for worse prognosis in men. However, office BP does not appropriately capture the real burden of hypertension in CKD, with ABP monitoring being recommended to obtain a more accurate risk stratification in this population [28]. Our findings confirm the value of this recommendation by showing that association between sex and either ESKD or mortality persists after adjustment of survival models for office BP. On the other hand, once the sex difference in ABP control is taken into account, men are no longer exposed to a significant increased risk of adverse outcomes, thus supporting the hypothesis that in CKD patients, sex-related difference in renal risk and mortality is explained, at least in part, by differences in ABP levels. Of note, this finding must not be interpreted as the need of a more intensive ABP control in men to reduce their renal risk, because such an information can be only supported by a randomized clinical trial. However, the lack of a significant interaction of ABP at goal and sex with ESKD and mortality risks suggests that the protective effects associated with controlled ABP do not differ between men and women.

No previous study has specifically addressed the impact of sex on the relationship between ABP and outcomes in non-dialysis CKD population. Individual-level meta-analysis of general population cohorts has shown that despite a lower absolute risk of cardiovascular events and all-cause death in women, the increase in risk associated with the 24-h and nighttime BP values is steeper in women than in men [29]. In that study, however, only 41% of patients were hypertensive (with the majority of them untreated) and no information on renal function was provided. Hermida *et al.* confirmed these findings in a mixed cohort of 1718 men and 1626 women with normotension, untreated hypertension or resistant hypertension [30]. In that study, mean eGFR was about 80 mL/min and less than one-fourth of patients were diagnosed with CKD.

Interestingly, the discrepancy between sexes in office and ABP control rate accounted for a higher prevalence of MUCH and SH in men versus women, two conditions associated with higher risk of progression of renal disease, death and cardiovascular events [10, 13, 14, 16]. Cross-sectional studies carried out in normotensive, untreated hypertensive or patients with resistant hypertension consistently report higher systolic and diastolic ABP values in men with a lower control rate of ABP [29–33]. However, the large majority of patients enrolled in those studies had normal renal function. Recently, CRIC investigators reported in a large cohort of CKD patients that altered BP profiles were associated with risk of adverse renal and cardiovascular outcomes and that the prognostic meaning of MUCH and SH was not modified by sex [10]. However, our data cannot be compared with those of the CRIC cohort because of several differences in the clinical characteristics of the population, such as ethnicity (only White in our study) and severity of CKD (lower eGFR and higher proteinuria in our study). More importantly, the sex-related difference in the ABP control rate, as well as ABP values in men and women, were not reported by CRIC investigators [10]. It is also important to note that women more frequently display white coat hypertension (Figure 1B). This finding further emphasizes the importance of ABP monitoring in order to detect women potentially exposed to the risk of cardio-renal hypoperfusion that may occur in patients with WCE when antihypertensive treatment is up-titrated on the basis of only office BP measures.

We are unable to clarify the exact mechanism(s) explaining higher ABP levels in men with CKD. Previous clinical and experimental studies have claimed sex differences in activation of RAS, sympathetic nervous activity and sex hormones [34–37]. We did not find any difference in the prevalence of uncontrolled ABP based on the use of either RAS inhibitors or beta-blockers; similarly, control rate of ABP did not differ in women based on age <50 or >50 years, the cut-off usually adopted to separate pre- and post-menopausal women [29]. However, the role of other critical factors involved in the different pathways of hypertension between men and women, such as endothelin-1, immune system, oxidative stress, inflammation and nitric oxide system [36, 37], cannot be assessed in our study either directly or indirectly.

The main limitation of this study is inherent to the observational nature of the study that cannot detect any cause–effect relationship. In addition, our data, obtained in White patients, cannot be generalized to other racial groups in which prevalence of elevated ABP and dipping status may differ [11]. Finally, our analysis is based on only one ABP monitoring, which may reduce the accuracy of risk stratification; however, diagnosis of abnormal 24-h BP profile is highly reproducible in CKD patients [38].

In conclusion, we provide novel evidence that in CKD patients under nephrology care: (i) men show higher daytime and nighttime BP levels than women, (ii) worse ABP control in men significantly contributes to the observed higher risk of ESKD and mortality, and (iii) sex-related differences in ABP levels and cardiorenal prognosis occur independently of office BP levels. Overall, these data support the importance of out-of-office BP monitoring to better refine the risk profiles of male and female CKD patients.

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AUTHORS' CONTRIBUTIONS

Research idea and study design were by R.M., F.B.G., L.D.N. and G.C. S.B., C.G., M.R., E.P. and V.B. were involved in data acquisition. R.M., F.B.G., R.A., L.D.N. and V.B. were involved in data analysis/interpretation. P.C., S.S., S.B. and C.G. were involved in statistical analysis. Supervision or mentorship carried out by R.M., F.B.G., R.A., L.D.N. and G.C. Each author contributed important intellectual content during manuscript drafting or revision, accepts personal accountability for the author's own contributions, and agrees to ensure that questions pertaining to the accuracy or integrity of any portion of the work are appropriately investigated and resolved.

CONFLICT OF INTEREST STATEMENT

R.M. has been member of Advisory Boards for Astellas and Amgen, and an invited speaker at meetings supported by Amgen and Vifor Pharma. R.A. has been: a member of data safety monitoring committees for AstraZeneca and Ironwood Pharmaceuticals; a member of steering committees of randomized trials for Akebia, Bayer, Janssen, GlaxoSmithKline, Reata, Relypsa, Sanofi and Genzyme US Companies; a member of adjudication committees for Bayer, Boehringer Ingelheim and Janssen; and a member of scientific advisory board or consultant for Celgene, Daiichi Sankyo, Inc., Eli Lilly, Relypsa, Reata, Takeda Pharmaceuticals, USA, ZS Pharma and Merck. E.P. has received consultant fees from Bayer, Vifor Fresenius and Novartis, and lecture fees from AstraZeneca, Vifor Pharma. M.R. has been a member of Advisory Boards for Astellas. L.D.N. has received fees for scientific consultation and/or lectures by Astellas, AstraZeneca, Mundibiopharma and Vifor Pharma. F.B.G., C.G., S.B., P.C., S.S., V.B. and G.C. have no conflicts to declare.

DATA AVAILABILITY STATEMENT

The data underlying this article will be shared on reasonable request to the corresponding author.

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